

| Question | Marking Guidance                                                                                                                                                                                                                                         | Mark                  | Comments                                                                                                                                                                                                                                                                                                      |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03.1     | 1. (Percentage of light absorbed at each wavelength) correlates (with rate of photosynthesis);<br>2. High(er)/increases (rate of photosynthesis) with blue <b>and</b> red light <b>and</b> low(er) /decreases (rate of photosynthesis) with green light; | 2<br>(2 x<br>AO3)     | 1. Accept description of correlates e.g. as % light absorbed increases so does photosynthesis or converse.<br>1. Ignore directly proportional.<br>2. Ignore wavelength.<br>2. Reject ' <b>no</b> photosynthesis with green light'.<br>2. Accept other words which are equivalent to 'high(er)' and 'low(er)'. |
| Question | Marking Guidance                                                                                                                                                                                                                                         | Mark                  | Comments                                                                                                                                                                                                                                                                                                      |
| 03.2     | 1. Intensity/brightness of light (at each wavelength)<br>2. Carbon dioxide concentration<br>3. Temperature<br>4. Water<br><b>OR</b>                                                                                                                      | 2 max<br>(2 x<br>AO2) | 3 correct = <b>2 marks</b><br>2 correct = <b>1 mark</b><br>0–1 correct = <b>0 marks</b><br>Ignore pH and nutrients.                                                                                                                                                                                           |

|          | Humidity;;                                                                                                      |                           | <p>1. Accept 'distance of/from light source'.</p> <p>1. Reject sunlight.</p> <p>2. Accept conc. for concentration.</p> <p>4. Accept water potential/<math>\Psi</math> (of soil).</p>                                                                                                                                                  |
|----------|-----------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question | Marking Guidance                                                                                                | Mark                      | Comments                                                                                                                                                                                                                                                                                                                              |
| 03.3     | <p>1. ATP and reduced NADP;</p> <p>2. ATP provides <u>energy</u>;</p> <p>3. GP reduced to triose phosphate;</p> | <p>3</p> <p>(3 x AO1)</p> | <p>1 Accept NADPH or NADPH<sub>2</sub> or NADP + H for reduced NADP.</p> <p>3. Must have idea of reduction. This may be conveyed by stating GP converted to triose phosphate using reduced NADP.</p> <p>3. Accept TP for triose phosphate as triose phosphate is named in the question.</p> <p>3. If GP named it must be correct.</p> |

